

Rohan Jain

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SUMMARY

MS Data Science graduate from CU Boulder (GPA 3.9/4.0, May 2026) with production experience designing ingestion pipelines and data-egress controls for an on-prem AI platform as an AI Engineering Intern at Bright Machines, plus ETL and geospatial data systems on AWS. Built data quality gates that cut failed pipeline runs by 90%, found a silent data-loss class in a production sync pipeline, and reverse-engineered proprietary binary formats for reproducible ingestion. Seeking Data Engineer roles focused on reliable pipelines and clean data architecture.

SKILLS

Data Engineering & Pipelines: ETL/ELT design, batch processing, incremental sync, change data capture (Debezium, Iceberg / Parquet), schema validation, data quality gates, data-egress controls (fail-closed allow-lists), pipeline cost modeling, vector index publishing

Databases & Storage: PostgreSQL, MySQL, MongoDB, PostGIS (spatial queries, buffer joins), Alembic migrations, SQLite / sqlite-vec, MinIO, Azure Blob, Databricks Unity Catalog; SQL (CTEs, window functions, indexing)

Languages & Frameworks: Python (pandas, NumPy, psycopg2, GeoPandas), Apache Spark, R, SQL, Bash, FastAPI, REST APIs; data serialization (JSON, XML, CSV, binary)

Cloud & Infrastructure: AWS (Lambda, S3, EC2, RDS, API Gateway), Azure (AKS, ACR, Blob), Kubernetes, Kustomize, ArgoCD, Docker, GitLab CI/CD, GitHub Actions; Temporal, Grafana, Loki, Prometheus

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Designed and shipped a read-only-by-construction **MCP** server from scratch (37 files, +3,163 lines, 12 tools) serving a 32,000-workflow **Temporal** namespace in production; Docker-packaged, deployed via **Kubernetes/Kustomize**.
- Built Confluence knowledge pipeline to run fully on-prem (**CQL** fetch in code, **Qwen** on an H100, no data egress) instead of per-page cloud LLM calls estimated at \$6/page, avoiding over **\$12K** of API spend across a 2,000+ page estate; recurring cost tracks ~12 changed pages/month.
- Found a silent data-loss class every check was blind to: the LLM merge step dropped up to 12% of grounded facts, including a remediation procedure, while passing all 30 tests.
- Shipped a 93-page, 9.9 MB vector index (bge-m3 embeddings, **sqlite-vec**) to production for other projects to consume, with a publish step that refuses when the build cannot prove its validity.
- Added a fail-closed Confluence space allow-list to an AI ingestion pipeline (102 to 85 pages), keeping customer-facing and personal spaces from ever leaving the network.
- Scoped AI adoption on the production **Databricks** lakehouse (Unity Catalog; Iceberg CDC from **Postgres** via Debezium); found Databricks AI had never been enabled, zero Model Serving endpoints, zero **Genie** spaces.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Reverse-engineered proprietary Thermo Isodat binary file formats (MFC CArchive serialization) by decompiling DLLs; implemented inheritance-aware deterministic R readers with fail-fast validation for 7+ instrument file types.
- Standardized stable isotope data ingestion across vendor formats: enforced consistent field names, units, schemas, and encoding to produce analysis-ready R tables with full data lineage.

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Built an end-to-end reproducible ML pipeline across 6 model types with schema, unit and range validation gates and versioned artifact bundles for auditable experiment comparison.
- Implemented an automated inference quality gate rejecting models whose post-ML MAE fails to beat the pre-ML baseline; reduced failed production runs 90% and saved 20 engineer-hours monthly.

EDUCATION

University of Colorado, Boulder

Aug 2024 – May 2026 | Boulder, CO

Master's in Data Science (GPA: 3.9/4.0)

Mumbai University, Rajiv Gandhi Institute of Technology

Jul 2020 – May 2024 | Mumbai, India

B.E. in Computer Engineering (GPA: 3.7/4.0)

PROJECTS

SafeRide - Geospatial Routing API | Python, FastAPI, PostgreSQL, PostGIS, AWS Lambda

Sep - Dec 2025

- Designed and implemented a REST API serving crash-aware route rankings; built ETL loading crash records and hazard data into PostGIS with composite risk scores via spatial buffer joins, deployed serverlessly on AWS Lambda with RDS.

Denver International Airport Analytics | Python, Power BI, Power Query

Oct 2024

- Integrated ServiceNow and Azure DevOps operational datasets via Python ETL; built dual interactive dashboards with drill-downs for SLA status and bottleneck detection, reducing manual reporting by 40%; placed 3rd.